

Project Proposal

Team Members

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Project Title: **Aetherion**

Type of Project: Web Application with AI-Powered Healthcare and Emergency Management Features

Problem Statements:

- Difficulty finding accurate hospital resource information: Medical records are scattered across providers, making it hard to access complete patient history
 - Time-consuming emergency response: Finding doctors, ambulances, blood donors, and medicines takes critical time during emergencies
 - Lack of continuous monitoring: No proper system for monitoring mothers and newborns after care
 - No centralized resource tracking: Difficulty tracking beds, ICU units, oxygen supply, and emergency services availability
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Objectives:

- To provide a centralized hospital information system that integrates all healthcare data
- To manage doctor directories and availability in real-time
- To track beds, ICU units, oxygen supply, and emergency services
- To enable appointment booking and management
- To store and manage patient medical records securely

- To provide AI-based healthcare assistance for preliminary diagnosis and recommendations
 - To support women, newborn care, and vaccination management
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Deliverables:

- User Management System (Patient, Doctor, Admin modules)
 - Hospital Resource Tracking System (beds, ICU, oxygen, emergency services)
 - Appointment Booking System
 - Pharmacy Management System
 - AI Health Assistant with recommendation engine
 - Women & Baby Care Management System
 - Emergency Services Module (ambulance, blood donor locator)
 - Administrative Dashboard for analytics and reporting
 - Technical documentation and user manuals
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| HTML | CSS | JavaScript | Node.js / PHP | MySQL |

Methodologies:

Waterfall: Traditional linear approach suitable for well-defined requirements, stable team, and fixed budget. Used for core infrastructure development.

Agile: Emphasizes collaboration, flexibility, and iterative development. Suitable for evolving AI features, dynamic team needs, and delivering working modules quickly. Selected as primary methodology for this project.

Project Overview:

Users' view: Users must create a profile by signing into the Smart Integrated Hospital & Emergency Management System. After login with email and password, they can search for doctors by specialization, location, or availability. Users can check real-time availability of hospital beds, ICU units, and oxygen supply. They can book appointments with doctors, access their medical history, and use the AI Health Assistant for symptom checking and health recommendations. Emergency features allow users to find nearby ambulances, blood donors, and pharmacies. Women can access pregnancy and baby care tracking, vaccination schedules, and newborn health monitoring.

Admin's view: Register/Login, Manage patient records, Manage doctor profiles and schedules, Track hospital resources (beds, ICU, oxygen), Monitor emergency service requests, Manage appointment bookings, Oversee pharmacy inventory, View AI assistant analytics, Manage women & baby care programs, Generate reports.

Steps:

1. User goes to the project web page
 2. User registers or logs into the system
 3. User can search for doctors by specialization or location
 4. User can check bed/ICU/oxygen availability in real-time
 5. User can book appointments with available doctors
 6. User can access and update medical records
 7. User can use AI Health Assistant for symptom checking
 8. User can request emergency services (ambulance, blood)
 9. Women users can access pregnancy and baby care modules
 10. User can view vaccination schedules and reminders
 11. User can order medicines from pharmacy management
 12. Admin can monitor all system activities
 13. Admin can update hospital resource status
 14. Admin can manage doctor schedules
 15. System generates analytics reports for admins
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Technology Stack:

- Frontend: HTML, CSS, JavaScript
 - Backend: Node.js / PHP
 - Database: MySQL
 - IDE: Visual Studio Code
 - AI Integration: Recommendation Engine / Machine Learning Model
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Risks and Dependencies:

- Internet connectivity dependency (critical for real-time updates)
- Data privacy and security challenges (patient health information)
- Hospital data integration requirements (API coordination needed)
- Third-party API availability (payment, SMS, maps)
- Healthcare data accuracy (requires validation mechanisms)

Future Scope:

- GPS tracking for ambulances and emergency services
- Telemedicine support for remote consultations
- Integration with wearable health devices
- Multi-language support
- Mobile application development